

UNIVERSITY COLLEGE LONDON

EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : **COMP0070**

ASSESSMENT PATTERN: **COMP0070A7PE 001**

MODULE NAME : **Algorithmics**

LEVEL : **Postgraduate**

DATE: : **28/04/2023**

TIME : **10:00**

DURATION : **2 Hours**

This paper is suitable for candidates who attended classes for this module in the following academic year(s):

**Year
2022-23**

**EXAMINATION PAPER CANNOT BE REMOVED FROM THE EXAM HALL.
PLACE EXAM PAPER AND ALL COMPLETED SCRIPTS INSIDE THE
EXAMINATION ENVELOPE**

Hall instructions	N/A
Additional materials	N/A
Standard Calculators	Yes
Non-Standard Calculators	No

TURN OVER

Answer BOTH questions.

Marks for each part of each question are indicated in square brackets.

Standard calculators are permitted.

1. a. Consider the following P matrix that has been returned by the FloydPath algorithm.

	A	B	C	D
A	0	0	B	B
B	D	0	0	0
C	0	0	0	0
D	0	A	B	0

- i. Decode the P matrix to give a connected graph that could have been the input to the FloydPath algorithm. Draw the graph and explain clearly how you obtained this from the P matrix.
- [6 marks]
- ii. Using the answer to part (i), assume that each arc has a weight (distance) of 1, and give the adjacency matrix that would be the input to Floyd's algorithm, and then generate each intermediate matrix and the final matrix that would be obtained with each iteration of Floyd's algorithm.
- [6 marks]
- b. For some undirected connected graphs, depth-first search and breadth-first search can result in identical search trees. Give a general class of graph for which this occurs and why?
- [6 marks]
- c. To calculate the binomial coefficient $\binom{n}{k}$ recursively, if $0 < k < n$, then $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$, otherwise $\binom{n}{k} = 1$. Give the pseudocode for a dynamic programming algorithm for calculating $\binom{n}{k}$ that only uses an array (not a matrix) to store the intermediate values.

[8 marks]

[Question 1 cont. over page]

- d. For the array $\langle 5, 8, 7, 6, 4, 3, 2, 1 \rangle$, consider a call to the Pivot algorithm with the first and last index positions for the array. Give each intermediate version of the array that results from an interchange of values, and make clear what values have been exchanged. Also give the value returned by the call to the pivot algorithm. Briefly, explain your answer.

[8 marks]

- e. For an undirected graph $G = (N, A)$, a **bridge** is an arc $(x, y) \in A$ such that there is no other path in G from x to y . Now consider the following statements where G is connected and $|N| = n$.

1. G has $n - 1$ arcs.
2. Each arc in G is a bridge.

Are the above statements equivalent? If yes, give a clear, precise, and concise explanation for why they are equivalent. If no, give a counterexample.

[8 marks]

- f. A binary tree can be encoded using three functions l , r and k such that for a node n , $l(n)$ gives the left child of n (or *nil* if there is none), $r(n)$ gives the right child of n (or *nil* if there is none), and $k(n)$ gives the key value at n . Let $\text{MaxKey}(l, r, x, k)$ be a simple recursive algorithm that takes a binary tree encoded by the l , r , and k functions, with the root x , and returns the maximum key value in the tree. Give the pseudocode for this algorithm.

[8 marks]

[Total for Question 1: 50 marks]

2. a. i. Explain what is meant by the term *elementary operation*. Why is the time complexity of an algorithm measured in terms of elementary operations? (5 marks)
- ii. Give the definition of the mathematical *order notation* $f(n) \in O(g(n))$, and explain how this concept relates to the algorithmic idea of *worst-case analysis*. (5 marks)

[10 marks]

- b. For each of the following statements, justify carefully, using the definition of big-O notation, whether they are true or false:

- i. $4n^5 + 10n^4 + 17 \in O(n^5)$ (3 marks);
- ii. $\log_3 n \in O(\log_2 n)$ (3 marks);
- iii. $4^n \in O(2^n)$ (4 marks).

[10 marks]

- c. For each of the following short pseudocode procedures, calculate the exact number of unit-time operations each procedure requires as a function of the input size n , and then simplify your answer using O -notation.

i. for $i = 1$ to n do
 for $j = 1$ to $2i+1$ do
 <unit cost operation>

(5 marks)

ii. for $i = 1$ to n do
 for $j = 1$ to n do
 for $k = 1$ to $(2i+j)$ do
 <unit cost operation>

(5 marks)

[10 marks]

[Question 2 cont. over page]

d. Solve the following recurrence relations and simplify your answer using O -notation.

You may assume the input n is a power of 2 where appropriate.

i.

$$f(0) = 2$$

$$f(n) = 5f(n-1) + 4 \quad \text{for } n \geq 1$$

(5 marks)

ii.

$$f(1) = 3$$

$$f(2) = 5$$

$$f(n) = 3f\left(\frac{n}{2}\right) - 2f\left(\frac{n}{4}\right) \quad \text{for } n > 2$$

(5 marks)

[10 marks]

e. i. Define the complexity class Co-NP. (2 marks)

ii. Prove that if $Y \leq_P X$ and $X \in \text{Co-NP}$ then $Y \in \text{Co-NP}$. (4 marks)

iii. Hence prove that if X is an NP-complete problem and $X \in \text{Co-NP}$ then $\text{NP} = \text{Co-NP}$. (4 marks)

[10 marks]

[Total for Question 2: 50 marks]

[Paper total: 100 marks]